

BUSINESS REQUIREMENTS

Data Cleaning & Validation

The raw table swiggy_data contains food delivery records across states, cities, restaurants, categories, and dishes.

The first goal is to ensure data quality by performing:

Null Check

Identify missing values in:

- State
- City
- Order_Date
- Restaurant_Name
- Location
- Category
- Dish_Name
- Price_INR
- Rating
- Rating_Count

SQL QUERY_1

```
SQL PROJECT...atha H (51) ✕
1  SELECT * FROM Swiggy_Data
2  -- Data Validation & Cleaning
3  --Null Check
4  SELECT
5      SUM(CASE WHEN State IS NULL THEN 1 ELSE 0 END) AS null_state,
6      SUM(CASE WHEN City IS NULL THEN 1 ELSE 0 END) AS null_state,
7      SUM(CASE WHEN Order_Date IS NULL THEN 1 ELSE 0 END) AS null_order_date,
8      SUM(CASE WHEN Restaurant_Name IS NULL THEN 1 ELSE 0 END) AS null_state,
9      SUM(CASE WHEN Location IS NULL THEN 1 ELSE 0 END) AS null_locatoin,
10     SUM(CASE WHEN Category IS NULL THEN 1 ELSE 0 END) AS null_Category,
11     SUM(CASE WHEN Dish_Name IS NULL THEN 1 ELSE 0 END) AS null_dish,
12     SUM(CASE WHEN Price_INR IS NULL THEN 1 ELSE 0 END) AS null_price,
13     SUM(CASE WHEN Rating IS NULL THEN 1 ELSE 0 END) AS null_rating,
14     SUM(CASE WHEN Rating_Count IS NULL THEN 1 ELSE 0 END) AS null_rating_count
15 FROM Swiggy_data;
```

RESULT-1

Results		Messages									
	null_state	null_state	null_order_date	null_state	null_location	null_Category	null_dish	null_price	null_rating	null_rating_count	
1	0	0	0	0	0	0	0	0	0	0	

SQL QUERY_2

Blank/Empty String Check

Detect fields containing blank values that may cause inaccurate analysis.

```
-- Blank or Empty Strigs
SELECT *
FROM Swiggy_Data
WHERE
State = '' OR City='' OR Restaurant_Name = '' OR Location='' OR Category = '' OR Dish_Name = ''
```

RESULT-2

Results		Messages									
	State	City	Order_Date	Restaurant_Name	Location	Category	Dish_Name	Price_INR	Rating	Rating_Count	

SQL QUERY_3

Duplicate Removal

Use ROW_NUMBER() to delete surplus duplicate rows while retaining one clean copy for each unique order.

```
--Duplicate Detection or Remove
SELECT
State, City, order_date, Restaurant_Name, location, Category, Dish_Name, Price_INR, rating, Rating_Count,
COUNT(*) AS CNT
FROM Swiggy_Data
GROUP BY State, City, order_date, Restaurant_Name, location, Category, Dish_Name, Price_INR, rating, Rating_Count
Having COUNT(*) > 1
```

RESULT-3

State	City	order_date	Restaurant_Name	location	Category	Dish_Name	Price_INR	rating	Rating_Count	CNT
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SQL QUERY_4

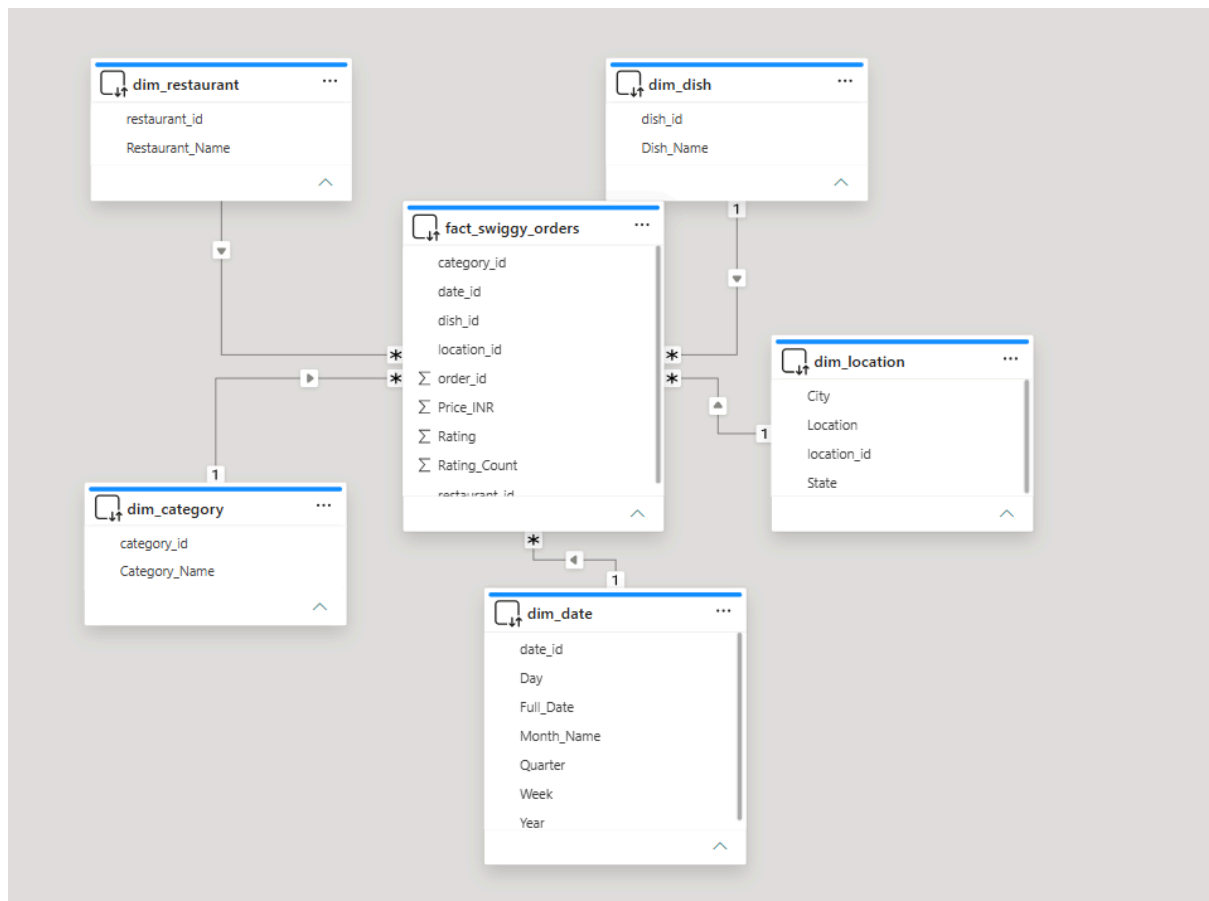
Dimensional Modelling (Star Schema)

To optimize analytics and reporting, build a **Star Schema** with the following dimension tables:

- **dim_date** → Year, Month, Quarter, Week
- **dim_location** → State, City, Location
- **dim_restaurant** → Restaurant_Name
- **dim_category** → Cuisine/Category
- **dim_dish** → Dish_Name

Central fact table:

- **fact_swiggy_orders** → Price_INR, Rating, Rating_Count, foreign keys to all dimensions



- dim_date Table

```
✓ -- CREATING SCHEMA
-- DIMENSION TABLES
-- DATE TABLE
✓ CREATE TABLE dim_date(
    date_id INT IDENTITY(1,1) PRIMARY KEY,
    Full_Date DATE,
    Year INT,
    MONTH INT,
    Month_Name VARCHAR(20),
    Quarter INT,
    Day INT,
    Week INT
)
DROP TABLE dim_date
```

Inserting data to dim_date Table

```
--INSERT THE DATA TO THE ALL THE TABLES
--dim_data
INSERT INTO dim_date (Full_Date, Year, Month_Name, Quarter, Day, Week)
SELECT DISTINCT
    order_date,
    YEAR(Order_Date),
    DATENAME(MONTH, Order_Date),
    DATEPART(QUARTER, Order_Date),
    DAY(Order_Date),
    DATEPART(WEEK, Order_Date)
FROM swiggy_data
WHERE Order_Date IS NOT NULL;
```

dim_location Table

```
--dim_location
CREATE TABLE dim_location (
    location_id INT IDENTITY(1,1) PRIMARY KEY,
    State VARCHAR(100),
    City VARCHAR(100),
    Location VARCHAR(200)
);
```

Inserting data to dim_location Table

```
-- dim_location
INSERT INTO dim_location (State, City, Location)
SELECT DISTINCT
    State,
    City,
    Location
FROM swiggy_data;
```

dim_restaurant

```
-- dim_restaurant
CREATE TABLE dim_restaurant (
    restaurant_id INT IDENTITY(1,1) PRIMARY KEY,
    Restaurant_Name VARCHAR(200)
);
```

Inserting data to dim_restaurant

```
-- dim_restaurant
INSERT INTO dim_restaurant(Restaurant_Name)
SELECT DISTINCT
    Restaurant_Name
FROM Swiggy_Data;
```

dim_category and dim_dish

```
-- dim_category
CREATE TABLE dim_category (
    category_id INT IDENTITY(1,1) PRIMARY KEY,
    Category_Name VARCHAR(200)
);

-- dim_dish
CREATE TABLE dim_dish(
    dish_id INT IDENTITY(1,1) PRIMARY KEY,
    Dish_Name VARCHAR(200)
);
```

Inserting data to dim_category and dim_dish

```
-- dim_category
INSERT INTO dim_category(Category_Name)
SELECT DISTINCT
    Category
FROM swiggy_data;

--dim_dish
INSERT INTO dim_dish(Dish_Name)
SELECT DISTINCT
    Dish_Name
FROM swiggy_data;
```

fact_table

-- CRETAE FACT TABLE

```
CREATE TABLE fact_swiggy_orders(  
    order_id INT IDENTITY (1,1) PRIMARY KEY,  
  
    date_id INT,  
    Price_INR DECIMAL(10,2),  
    Rating DECIMAL (4,2),  
    Rating_Count INT,  
  
    location_id INT,  
    restaurant_id INT,  
    category_id INT,  
    dish_id INT,  
  
    FOREIGN KEY (date_id) REFERENCES dim_date(date_id),  
    FOREIGN KEY (location_id) REFERENCES dim_location(location_id),  
    FOREIGN KEY (restaurant_id) REFERENCES dim_restaurant(restaurant_id),  
    FOREIGN KEY (category_id) REFERENCES dim_category(category_id),  
    FOREIGN KEY (dish_id) REFERENCES dim_dish(dish_id)  
);
```

Inserting to the fact_table

```
-- Fact table

INSERT INTO fact_swiggy_orders
(
    date_id,
    Price_INR,
    Rating,
    Rating_Count,
    location_id,
    restaurant_id,
    category_id,
    dish_id
)
SELECT
    dd.date_id,
    s.Price_INR,
    s.Rating,
    s.Rating_Count,
    dl.location_id,
    dr.restaurant_id,
    dc.category_id,
    dsh.dish_id

FROM swiggy_data s

JOIN dim_date dd
    ON dd.Full_Date = s.Order_Date

JOIN dim_location dl
    ON dl.State = s.State
    AND dl.City = s.City
    AND dl.Location = s.Location

JOIN dim_restaurant dr
    ON dr.Restaurant_Name = s.Restaurant_Name

JOIN dim_category dc
    ON dc.Category_Name = s.Category

JOIN dim_dish dsh
    ON dsh.Dish_Name = s.Dish_Name;

-- data in all combine version

SELECT * FROM fact_swiggy_orders f
JOIN dim_date d
    ON f.date_id = d.date_id
JOIN dim_location l
    ON f.location_id = l.location_id
JOIN dim_restaurant r
    ON f.restaurant_id = r.restaurant_id
JOIN dim_category c
    ON f.category_id = c.category_id
JOIN dim_dish dl
    ON f.dish_id = dl.dish_id;
```

KPI

Total Orders

```
-- Total Orders
SELECT COUNT(*) AS Total_Orders
FROM fact_swiggy_orders|
```

Result

Results		Messages	
	Total_Orders		
1	197401		

Total Revenue

```
-- Total Revenue (INR Million)
SELECT FORMAT(SUM(CONVERT(FLOAT ,price_INR))/1000000,'N2') + ' INR Million' AS Toatal_Revenue
FROM fact_swiggy_orders
```

Result

Results		Messages	
	Toatal_Revenue		
1	53.00 INR Million		

Average Dish Price

```
--Average Dish Price
SELECT FORMAT(AVG(CONVERT(FLOAT ,price_INR)),'N2') + ' INR ' AS Toatal_Revenue
FROM fact_swiggy_orders
```

Result

Results		Messages	
	Toatal_Revenue		
1	268.50 INR		

Average Rating

```
--Average Rating
SELECT
AVG(Rating) AS Avg_Raing
FROM fact_swiggy_orders
```

Result

Results Messages	
	Avg_Raing
1	4.341577

-Deep-Dive Busines Analysis

Monthly Order Trends

```
-- Monthly Order Trends
SELECT
d.Year,
d.month_name,
COUNT(*) AS Total_orders
FROM fact_swiggy_orders f
JOIN dim_date d
ON f.date_id = d.date_id
GROUP BY d.Year,
d.month_name
ORDER BY COUNT(*) DESC
```

Result

Results Messages			
	Year	month_name	Total_orders
1	2025	January	25393
2	2025	August	25227
3	2025	May	25188
4	2025	July	24936
5	2025	April	24584
6	2025	March	24400
7	2025	June	24382
8	2025	February	23291

-- Monthly Total_Revenue

```
-- Monthly Total_Revenue
SELECT
  d.Year,
  d.month_name,
  FORMAT(SUM(CONVERT(FLOAT, price_INR))/1000000, 'N2')+ ' INR Million' AS Total_orders
FROM fact_swiggy_orders f
JOIN dim_date d
ON f.date_id = d.date_id
GROUP BY d.Year,
d.month_name
ORDER BY Total_orders DESC
```

Result

	Year	month_name	Total_orders
1	2025	January	6.82 INR Million
2	2025	August	6.79 INR Million
3	2025	May	6.79 INR Million
4	2025	July	6.65 INR Million
5	2025	April	6.59 INR Million
6	2025	March	6.57 INR Million
7	2025	June	6.51 INR Million
8	2025	February	6.27 INR Million

-- Orders by Day of Week (mon-Sun)

```
-- Orders by Day of Week (mon-Sun)
SELECT
  DATENAME(WEEKDAY, d.full_date) AS day_name,
  COUNT(*) AS total_orders
FROM fact_swiggy_orders f
JOIN dim_date d ON f.date_id = d.date_id
GROUP BY
  DATENAME(WEEKDAY, d.full_date),
  DATEPART(WEEKDAY, d.full_date)
ORDER BY
  DATEPART(WEEKDAY, d.full_date) DESC;
```

Result

	day_name	total_orders
1	Saturday	28933
2	Friday	28284
3	Thursday	28450
4	Wednesday	28284
5	Tuesday	27413
6	Monday	27568
7	Sunday	28469

Location Based Analysis

Top 10 cities by order volume

```
-- Top 10 cities by order volume
SELECT TOP 10
l.City,
COUNT(*) AS Total_Orders
FROM fact_swiggy_orders f
JOIN dim_location l
ON l.location_id = f.location_id
GROUP BY l.City
ORDER BY COUNT(*) DESC
```

Result

	City	Total_Orders
1	Bengaluru	20072
2	Mumbai	10507
3	Hyderabad	10308
4	Jaipur	10285
5	Lucknow	10192
6	New Delhi	10191
7	Ahmedabad	10175
8	Chandigarh	10060
9	Kolkata	10044
10	Chennai	10042

Revenue contribution by states

```
-- Revenue contribution by states
SELECT TOP 10
l.State,
SUM(f.price_INR) AS Total_Revenue
FROM fact_swiggy_orders f
JOIN dim_location l
ON l.location_id = f.location_id
GROUP BY l.State
ORDER BY SUM(f.price_INR) DESC
```

Result

	State	Total_Revenue
1	Karnataka	5455887.73
2	Uttar Pradesh	3117359.65
3	Telangana	3021656.62
4	Maharashtra	3015573.35
5	Delhi	2829180.60
6	Gujarat	2815536.27
7	Punjab	2804991.82
8	West Bengal	2662213.76
9	Tamil Nadu	2642594.63
10	Rajasthan	2502833.61

Food Performance

Top 10 restaurants by orders

-- Top 10 restaurants by orders

```
SELECT TOP 10
r.restaurant_name,
COUNT(*) AS Total_Orders
FROM fact_swiggy_orders f
JOIN dim_restaurant r
ON r.restaurant_id = f.restaurant_id
GROUP BY r.restaurant_name
ORDER BY COUNT(*) DESC
```

Result

	restaurant_name	Total_Orders
1	McDonald's	13528
2	KFC	12957
3	Burger King	7115
4	Pizza Hut	6529
5	Domino's Pizza	5489
6	LunchBox - Meals and Thalís	4700
7	Baskin Robbins - Ice Cream Desserts	4197
8	Faasos - Wraps, Rolls & Shawarma	3256
9	Olio - The Wood Fired Pizzeria	3239
10	The Good Bowl	2665

Most Orderd dish

```
-- Most Orderd dish
SELECT TOP 10
d.Dish_Name,
COUNT(*) AS Order_Count
FROM fact_swiggy_orders f
JOIN dim_dish d
ON d.dish_id = f.dish_id
GROUP BY d.Dish_Name
ORDER BY Order_Count DESC
```

Result

	Dish_Name	Order_Count
1	Veg Fried Rice	321
2	Choco Lava Cake	303
3	Jeera Rice	265
4	Paneer Butter Masala	262
5	French Fries	248
6	Chicken Sausage	230
7	Chicken Fried Rice	228
8	Butter Naan	218
9	Margherita Pizza	203
10	Green Salad	197

Total Orders by Price Range

```
-- Total Orders by Price Range
SELECT
CASE
    WHEN CONVERT(FLOAT, price_inr) < 100 THEN 'Under 100'
    WHEN CONVERT(FLOAT, price_inr) BETWEEN 100 AND 199 THEN '100 - 199'
    WHEN CONVERT(FLOAT, price_inr) BETWEEN 200 AND 299 THEN '200 - 299'
    WHEN CONVERT(FLOAT, price_inr) BETWEEN 300 AND 499 THEN '300 - 499'
    ELSE '500+'
END AS price_range,
COUNT(*) AS total_orders
FROM fact_swiggy_orders
GROUP BY
CASE
    WHEN CONVERT(FLOAT, price_inr) < 100 THEN 'Under 100'
    WHEN CONVERT(FLOAT, price_inr) BETWEEN 100 AND 199 THEN '100 - 199'
    WHEN CONVERT(FLOAT, price_inr) BETWEEN 200 AND 299 THEN '200 - 299'
    WHEN CONVERT(FLOAT, price_inr) BETWEEN 300 AND 499 THEN '300 - 499'
    ELSE '500+'
END
ORDER BY total_orders DESC;
```

Result

	price_range	total_orders
1	100 - 199	56189
2	200 - 299	54567
3	300 - 499	43758
4	Under 100	26795
5	500+	16092

Top 10 Rating Count Distribution

```
--Rating Count Distribution
SELECT
    rating,
    COUNT(*) AS rating_count
FROM fact_swiggy_orders
GROUP BY rating
ORDER BY COUNT(*) DESC;
```

Result

	rating	rating_count
1	4.40	85642
2	4.30	13698
3	4.60	10840
4	4.50	9946
5	5.00	9401
6	4.70	9089
7	4.80	8809
8	4.20	8214
9	4.10	7619
10	4.90	5713

IMP KPI'S

TOTAL ORDERS : 197,401

TOTAL REVENUE : ₹53.00 Million

AVERAGE DISH PRICE : ₹268.50

AVERAGE RATING : 4.341577

TOP CITY : Bengaluru — 20,072 orders

TOP STATE REVENUE : Karnataka — ₹5,455,887.73

TOP RESTAURANT : McDonald's — 13,528 orders

TOP DISH : Veg Fried Rice — 321 orders

TOP PRICE RANGE : ₹100–₹199 — 56,189 orders

MOST COMMON RATING : 4.40 — 85,642 orders